SMILES-Alternative

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1 SMILES Alternatives

1.1 SELFIES

1.1.1 Pros

- SELFIES is a robust version of SMILES. Therefore, every SELFIES is grammatically valid unlike SMILES which can be invalid. This is applicable for encoder-only transformers during SELFIES augmentation, since alternative representations of the same compound can be generated using SELFIES with 100% accuracy
- SMILES is easily transferable to SELFIES using the selfies library
- SELFIES uses a *length-encoding system*. The length-encoding system is the process in which an index is placed before every branch/part of the compound. This can potentially lead to better model understanding.

1.1.2 Cons

- SELFIES strings are often 2-3 times longer than SMILES strings, leading to higher computational cost and potential memory errors with tokenization length.

1.1.3 Transformation from SMILES to SELFIES

SMILES can be transformed to SELFIES using the selfies library on GitHub

(GitHub Link: https://github.com/aspuru-guzik-group/selfies)

1.1.4 Pre-Trained Model

Name: SELFormer

- **Source**: GitHub:https://github.com/HUBioDataLab/SELFormer?tab=readme-ov-file
- **Architecture**: Built on the RoBERTa transformer architecture, which is an Encoder-only Transformer.
- **Pre-Trained Dataset Size**: 2.08 million SELFIES strings

Name: SELFIES-TED

- **Source**: HuggingFace: https://huggingface.co/ibm-research/materials.selfies-ted
- **Architecture**: Built on BART model architecture, which is an Encoder-Decoder Transformer

- **Pre-Trained Dataset Size**: 1 billion SELFIES Strings

Name: MolGen

- **Source**: HuggingFace: https://huggingface.co/zjunlp/MolGen-large, GitHub: https://github.com/zjunlp/MolGen
- **Architecture**: Built on the BART model architecture, which is an Encoder-Decoder Transformer
- **Pre-Trained Dataset Size**: 100+ million SELFIES Strings

1.2 Molecular Graphs

1.2.1 Pros

- Molecular graphs are definite representations of compounds, and no permutations can be generated for the same compound. Therefore, the model doesn't have to learn different representations of the same compound, unlike SMILES-trained models.
- Each node (atom) and edge (bond) in a molecular graph can contain additional molecular information, such as bond energy or electronegativity.
- SMILES can be converted to Molecular Graphs using RDKit

1.2.2 Cons

- Each molecular graph is computationally expensive compared to a SMILES string, leading to longer training times
- The additional molecular information contained in each node or edge can also cause information overload, resulting in a decrease in model performance

1.2.3 Transformation from SMILES to Molecular Graphs

SMILES can be transformed into Molecular Graphs using RDKit's library

1.2.4 Pre-Trained Model

Name: Graphormer

- **Source**: GitHub: https://github.com/microsoft/Graphormer
- **Architecture**: Transformer model with structural encodings specialized for graph-structured data
- **Pre-Trained Dataset Size**: 3.8 million molecules

Name: GraphBRAIN

- **Source**: GitHub: https://github.com/Rishit-dagli/GraphBRAIN/blob/main/model/training/train.py, Paper: https://arxiv.org/html/2409.04909v1
- **Architecture**: GPS++ block (MPNN variant), MPNN block (Message Passing Neural Networks that aggregate information from neighboring nodes and edges to update node features), Standard Attention Block, Feed Forward Network

- **Pre-Trained Dataset Size**: 3.7 million molecules

Name: UniMol

- **Source**: GitHub: https://github.com/deepmodeling/Uni-Mol
- **Architecture**: Transformer-based model with a 3D Positional Encoding and Self-Attention for pair representations
- **Pre-Trained Dataset Size**: 209 million 3D conformations